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INVITED TALK / ABSTRACT

Exponential growth of commensurability classes of almost totally geodesic surface subgroups in hyperbolic 3-manifolds

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Abstract

In this talk we will discuss how to use Kahn-Markovic surface construction to show that for any given error threshold, the number of incommensurable almost totally geodesic surface subgroups of a finite volume hyperbolic 3-manifold grows exponentially on the genus of the surface. As an application, we will discuss the incommensurable version of minimal surface entropy. Based on on-going work with Fernando Al-Assal and Jia Wan.

Keywords: commensurability classes; almost totally geodesic surface subgroups; hyperbolic 3-manifolds; Kahn-Markovic surface construction; incommensurable surface subgroups; genus; minimal surface entropy; exponential growth